

Jinuk Moon

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Seoul National University

PROFILE

PhD candidate developing [machine learning interatomic potentials for accelerating catalyst discovery](#). Developed [CatBench](#), a systematic benchmarking framework that enables reliable evaluation of MLIPs in heterogeneous catalysis. Research focuses on [combining machine learning with first-principles calculations](#) to discover efficient catalytic materials for clean energy applications.

EDUCATION

- Seoul National University** 2023 – Present
Ph.D. Materials Science and Engineering Seoul, South Korea
 - Machine learning-accelerated computational catalysis for electrocatalyst discovery and design
- Pohang University of Science and Technology (POSTECH)** 2019 – 2022
B.S. Chemical Engineering Pohang, South Korea
 - Thesis: Study of Metal-Nitrogen-doped carbon catalysts for Chlorine Evolution Reaction combining DFT calculation with Machine Learning
 - GPA: 3.98/4.3 (Summa Cum Laude)
- KU Leuven** Spring 2022
Exchange Student - Faculty of Science Leuven, Belgium

SOFTWARE & OPEN SOURCE

- CatBench - Benchmark Framework for Machine Learning Interatomic Potentials in Catalysis**
Systematic evaluation framework for MLIPs | Pip-installable | catbench.org | github.com/JinukMoon/CatBench
- ensemble-GA - Ensemble Genetic Algorithm for Structure Optimization**
Efficient genetic algorithm using ensemble strategy for catalyst discovery | github.com/JinukMoon/ensemble-GA
- alloy-catalyst-atlas - MLIP-Accelerated High-Throughput Catalyst Screening Atlas**
Systematic mapping of ternary alloy surfaces for electrocatalyst discovery | github.com/JinukMoon/alloy-catalyst-atlas

HONORS & AWARDS

- Presidential Science Scholarship (PhD Track)** 2025
Ministry of Science and ICT South Korea
- Best Oral Award** Spring 2025
The Korean Institute of Metals and Materials (KIM) South Korea
 - CatBench: Benchmark of Machine Learning Interatomic Potentials for Adsorption Energy Predictions in Heterogeneous Catalysis
- Best Oral Award** Fall 2024
The Korean Institute of Metals and Materials (KIM) South Korea
 - Finding optimal MoC catalyst for hydrogen evolution via fine-tuned machine learning potential
- Best Oral Award** Spring 2024
The Korean Ceramic Society (KCerS) South Korea
 - Selective control of oxygen evolution mechanisms by doping concentrations in RuO₂

SKILLS

- Programming Language:** Python (Advanced), PyTorch, C/C++ (Intermediate), Bash scripting (Intermediate), HTML
- Machine Learning:** MLIP Training/Fine-tuning, MLIP Benchmarking, Genetic Algorithm, Dataset Generation, Bayesian Optimization, ML Analysis of Experimental Characterization
- Computational Method:** Density Functional Theory, Machine Learning Interatomic Potentials (experience with 15+ models including UMA, EquiformerV2, GemNet-OC, Equiformer, Matlantis, MACE), Molecular Dynamics, Electronic Structure Analysis
- Server Management:** Supercomputer Cluster Server Management, SLURM, PBS, Batch Job Optimization
- Research Expertise:** Heterogeneous Catalysis, Single Atom Catalysis, Electrocatalysis (OER, HER, CER), Multiscale Simulation using MLIP, MLIP Benchmarking
- Language:** Korean (Native), English (Fluent)

[†] Equal contribution, * Corresponding author

- [Submitted] Jun, H.[†], Kwon, D.[†], **Moon, J.**[†], Han, S.[†], Kim, S., Yi, S. Y., Kim, Y., Kang, E., Choi, C. H.* , Han, J. W.* , Kim, W.* , Lee, J.* (2025). Synergistic Effect of Tuning Reaction Mechanism and Interfacial Water for Efficient Acidic Water Oxidation. *Submitted*.
- [revision] Park, K.[†], Lee, W.[†], Park, J., Jo, M., Bae, Y., Kim, M., Shin, S., **Moon, J.**, O'Hayre, R.* , Han, J. W.* , Song, S.* , Park, J. Y.* (2025). Breakthrough oxygen electrode reaction kinetics using BaCe_{0.7}Zr_{0.1}Y_{0.1}Yb_{0.1}O_{3-δ} based composite air electrodes for reversible solid oxide cells. *Nature Energy*, in revision.
- [1] Choung, S.[†], Kim, M.[†], **Moon, J.**[†], Han, J. W.* (2025). From Atomic Motif to Realistic Single Atom Catalysts through Machine Learning Interatomic Potentials. *ACS Energy Letters*, 10, 6288-6296.
- [2] **Moon, J.**, Jeon, U., Choung, S., Han, J. W.* (2025). CatBench Framework for Benchmarking Machine Learning Interatomic Potentials in Adsorption Energy Predictions for Heterogeneous Catalysis. *Cell Reports Physical Science*, 6, 102968.
- [3] Jung, H.[†], Lee, Y.[†], Cho, K.[†], Kim, T. Y., Song, J., Jung, H. S., Park, S. I., Lee, Y., **Moon, J.**, Park, W., Nam, J., Park, S., Kim, W., Han, J. W.* (2025). Strategic Design of Oxophilic Dopants for Active and Durable Alkaline Hydrogen Evolution Reaction Under Seawater. *Advanced Functional Materials*, e16387.
- [4] Jun, H.[†], Kang, E.[†], **Moon, J.**[†], Kim, H., Han, S., Choung, S., Kim, S., Yi, S. Y., Kang, E., Choi, C. H.* , Han, J. W.* , Lee, J.* (2025). Quantity effect of heteroatom incorporation on the oxygen evolution mechanism in ruthenium oxide. *Chem*, 11(5), 102367.
- [5] Choung, S.[†], Park, W.[†], **Moon, J.**[†], Han, J. W.* (2024). Rise of machine learning potentials in heterogeneous catalysis: Developments, applications, and prospects. *Chemical Engineering Journal*, 494, 152757.
- [6] Choung, S.[†], Yang, H.[†], **Moon, J.**, Park, W., June, H., Lim, C., Han, J. W.* (2024). Theoretical tuning of local coordination environment of metal-nitrogen doped carbon catalysts for selective chlorine-evolution reaction. *Catalysis Today*, 425, 114358.
- [7] Choung, S., Kim, Y., **Moon, J.**, Roh, J., Hwang, J., Han, J. W.* (2023). Unveiling the catalyst deactivation mechanism in the non-oxidative dehydrogenation of light alkanes on Rh(111): Density functional theory and kinetic Monte Carlo study. *Catalysis Today*, 411-412, 113819.

CONFERENCE PRESENTATIONS

- [1] **Moon, J.** (2025). Machine Learning-Accelerated Multiscale Modeling of Metal/MoC Electrocatalysts for Alkaline Hydrogen Evolution (Poster). *The 8th International Conference on Advanced Electromaterials (ICAE 2025)*, Jeju, South Korea.
- [2] **Moon, J.** (2025). Machine Learning-Accelerated Multiscale Modeling of Metal/MoC Electrocatalysts for Alkaline Hydrogen Evolution (Poster). *Korean Institute of Metals and Materials (KIM)*, Gwangju, South Korea.
- [3] **Moon, J.** (2025). Multiscale Modeling of Nanostructured Electrocatalyst for Alkaline Water Electrolysis via Fine-tuned Machine Learning Interatomic Potential (Oral Presentation). *Korean Institute of Chemical Engineers (KICHE)*, Jeju, South Korea.
- [4] **Moon, J.** (2025). CatBench: Benchmark Framework of Machine Learning Interatomic Potentials for Adsorption Energy Predictions in Heterogeneous Catalysis (Poster). *The 29th North American Catalysis Society Meeting (NAM29)*, Atlanta, United States.
- [5] **Moon, J.** (2025). Multiscale Modeling of Nanostructured Electrocatalyst for Alkaline Water Electrolysis via Graph Neural Network (Oral Presentation). *The 29th North American Catalysis Society Meeting (NAM29)*, Atlanta, United States.
- [6] **Moon, J.** (2025). CatBench: Benchmark Framework of Machine Learning Interatomic Potentials for Adsorption Energy Predictions in Heterogeneous Catalysis (Poster). *The Korean Society of Industrial and Engineering Chemistry (KSIEC)*, Jeju, South Korea.
- [7] **Moon, J.** (2025). CatBench: Benchmark of Machine Learning Interatomic Potentials for Adsorption Energy Predictions in Heterogeneous Catalysis (Oral Presentation). *Korean Institute of Metals and Materials (KIM)*, Jeju, South Korea.
- [8] **Moon, J.** (2024). Finding optimal MoC catalyst for hydrogen evolution via fine-tuned machine learning potential (Oral Presentation). *Korean Institute of Metals and Materials (KIM)*, PyeongChang, South Korea.
- [9] **Moon, J.** (2024). Finding optimal MoC catalyst for hydrogen evolution via fine-tuned machine learning potential (Oral Presentation). *23rd Korea Japan Students' Symposium*, Seoul, South Korea.
- [10] **Moon, J.** (2024). Selective Control of Oxygen Evolution Reaction Mechanisms by Adjusting Mn Doping Concentrations in RuO₂ (Oral Presentation). *The Korean Ceramic Society (KCerS)*, Busan, South Korea.

TEACHING EXPERIENCE

- **Seoul National University**
Teaching Assistant - Introduction to Computer Programming (Python)
Fall 2025
Seoul, South Korea
- **Seoul National University**
Teaching Assistant - Design of Catalytic Materials via Computational Materials Science and ML
Fall 2024
Seoul, South Korea
- **Seoul National University**
Teaching Assistant - Physical Chemistry
Fall 2023
Seoul, South Korea